

Experiment - 7.1.2. Possible Combinations from Three Digits

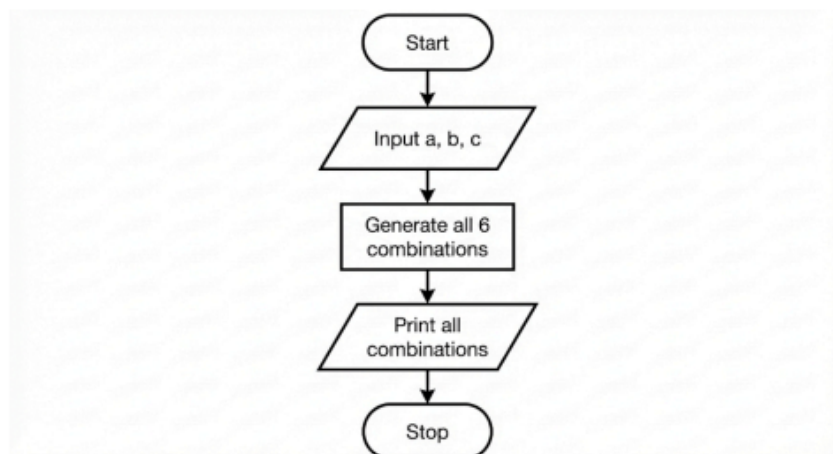
1. Aim

To design and implement a Python program that accepts three single-digit inputs (separated by spaces) and prints all six possible 3-digit combinations of these digits. The program must treat the inputs as strings to maintain any leading zeros through concatenation.

2. Pseudocode

1. **START**
2. **READ** a single line of input from the user, split it by spaces, and **STORE** the three string values into variables a, b, and c.
3. **PRINT** the concatenation of a, b, and c (representing the combination abc).
4. **PRINT** the concatenation of a, c, and b (representing the combination acb).
5. **PRINT** the concatenation of b, a, and c (representing the combination bac).
6. **PRINT** the concatenation of b, c, and a (representing the combination bca).
7. **PRINT** the concatenation of c, a, and b (representing the combination cab).
8. **PRINT** the concatenation of c, b, and a (representing the combination cba).
9. **END**

3. Flowchart



7.1.2. Possible Combinations from Three Digits

Write a Python program to accept three digits and print all possible 3-digit combinations that can be formed using those three digits. Each digit should be used exactly once in each combination.

For three digits a , b , and c , there are 6 possible arrangements based on their indices: abc , acb , bac , cab , cba , cba .

Input Format:

- Single line contains three space-separated non-negative integers, a , b , and c , representing the three digits

Output Format:

- Print all possible 3-digit combinations
- Each combination should be printed as a 3-digit number on a separate line
- Print combinations as they appear (maintain leading zeros if first digit is 0)

Constraints:

- $0 \leq \text{each digit} \leq 9$
- Digits can be same or different

Sample Test Cases

combinat...

```

1 a, b, c = input().split()
2 digits = [a, b, c]
3
4 for i in range(3):
5     for j in range(3):
6         for k in range(3):
7             if i != j and i != k and j != k:
8                 print(digits[i] + digits[j] + digits[k])

```

Average time

0.003 s

2.86 ms

Maximum time

0.004 s

4.00 ms

3 out of 3 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 4 ms

Debug

Expected output

1 2 3
123
132
213
231
312

Actual output

1 2 3
123
132
213
231
312

Terminal

Test cases

< Prev

Reset

Submit

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